

niBuild: Towards FAIR multi-modal neuroimaging analysis

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[Demo]



[Docs]

Motivation

Recent advances in neuroimaging have led to an **unprecedented emphasis on imaging** in brain research. Despite progress in standardizing data dissemination, **analysis remains fragmented**.

The absence of analysis workflow standards leads to **irreproducible** and potentially **invalid results**, **opacity** in the literature, and **reinforces archaic** analysis practices.

Core Contribution

niBuild is a **web-based graphical user interface (GUI)** enabling **visual** construction of multi-modal preprocessing and analysis workflows.

The open-source architecture follows the **FAIR principles** for optimization of data sharing. niBuild workflows are **portable, environment-agnostic containers**. Individual operations are implemented via the **Common Workflow Language (CWL)** as standalone tools.

Library	Tools	Modality	Tools
FSL	52	fMRI	60
AFNI	47	sMRI	46
FreeSurfer	23	Utility	36
Pipelines*	20	dMRI	24
ANTs	17	ASL	4
MRtrix3	8	PET	1
Connectome Workbench	5	Multimodal	1

Table 1. niBuild implemented tool counts. Includes canonical tools from popular libraries with coverage across imaging modalities. CWL-based architecture enables extension of tool offerings. Pipelines* are pre-built workflows as single composite nodes (e.g., fMRIprep).

Interface & Usage

Figure 1. Workflow builder interface displaying an fMRI workflow.

- Workflow menu:** list of 172 tools and pipelines, plus I/O and saved workflows.
- Workflow canvas:** drag-and-drop workflow construction space.
- CWL preview:** live CWL workflow and job YAML generated during construction.
- Utility console:** Live validation, logging, I/O summary, environment variable access, and server connectivity.

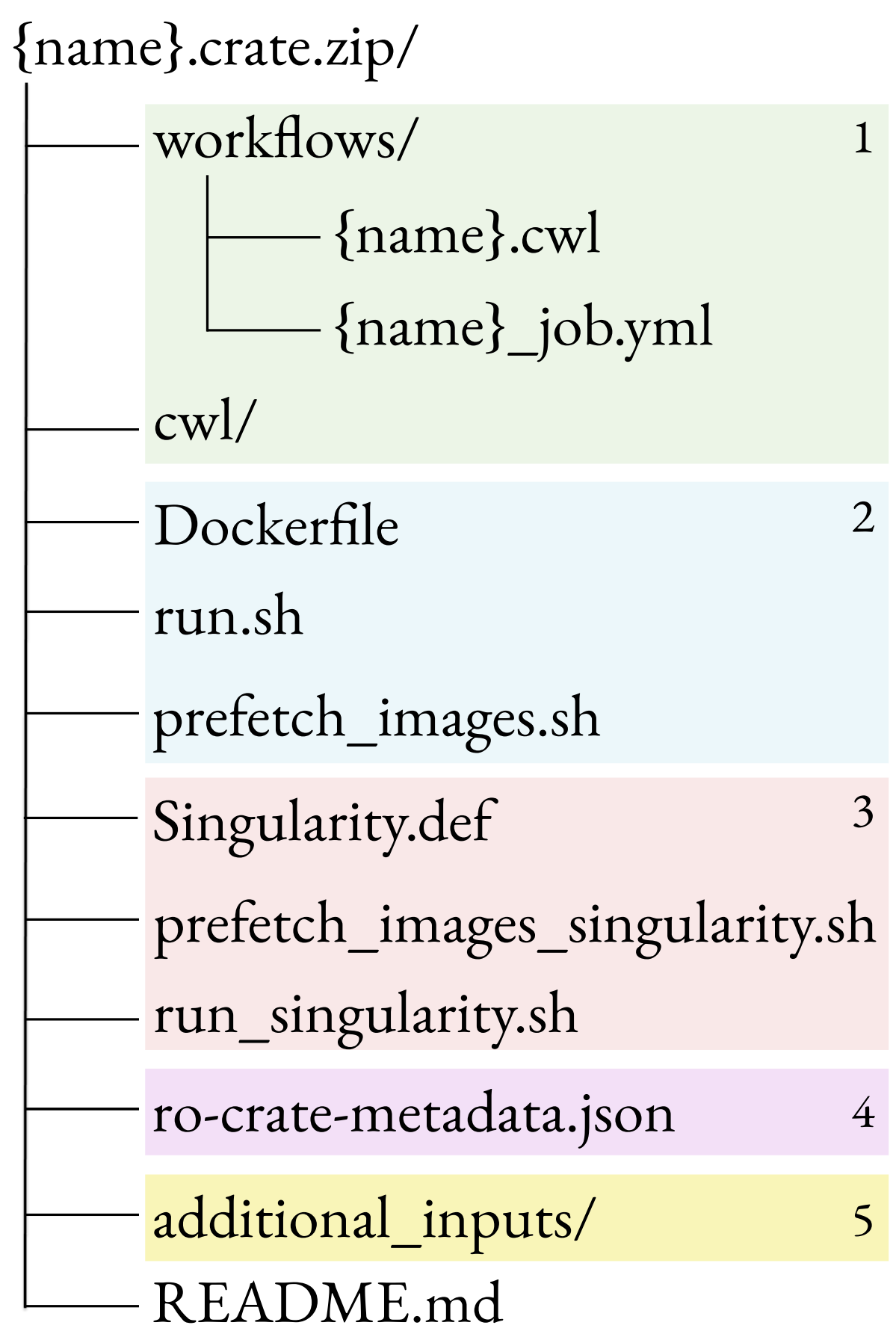
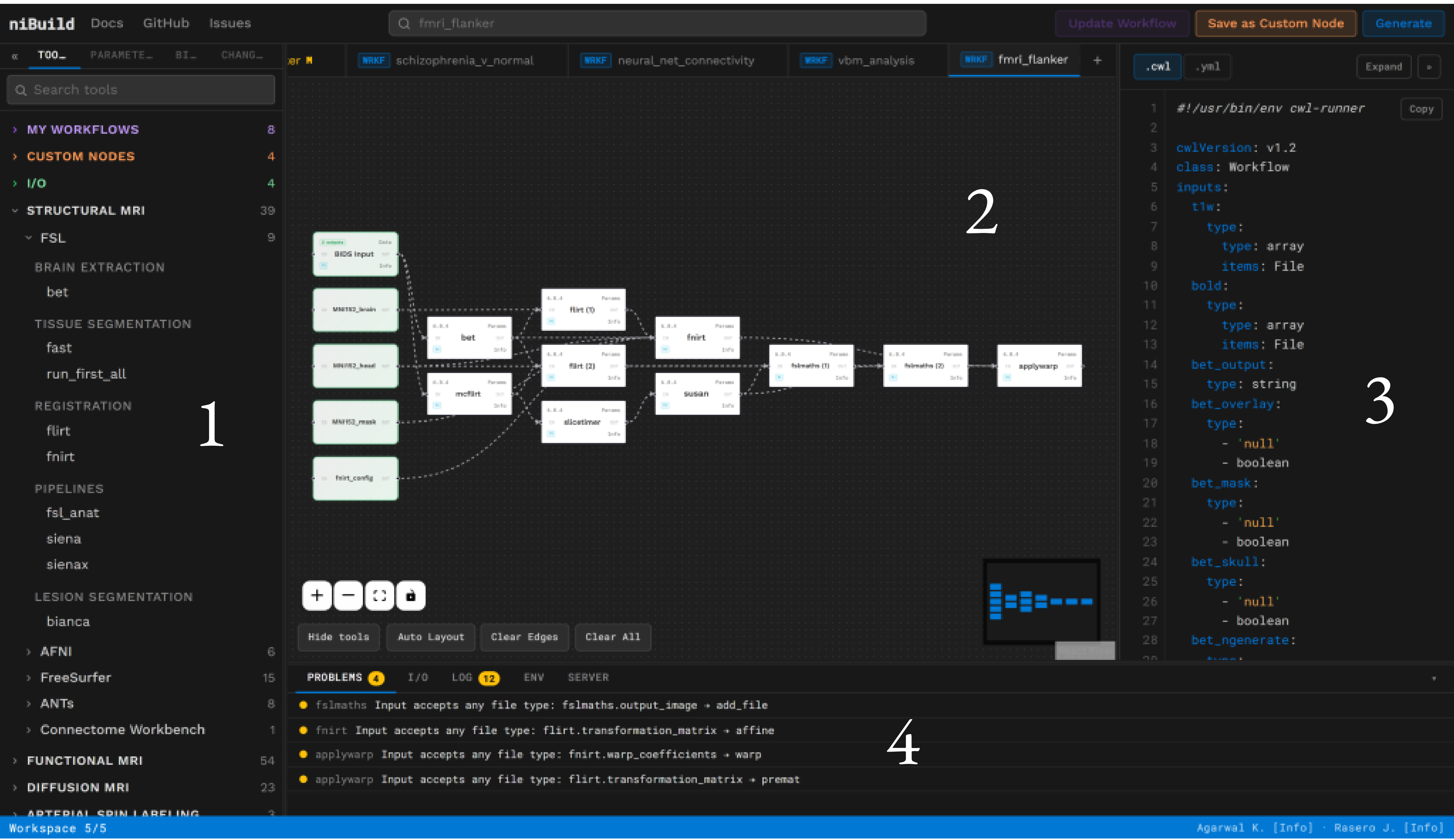


Figure 2. Exported Workflow RO-Crate bundle. Compilation of source and container files, metadata, and documentation.

Results

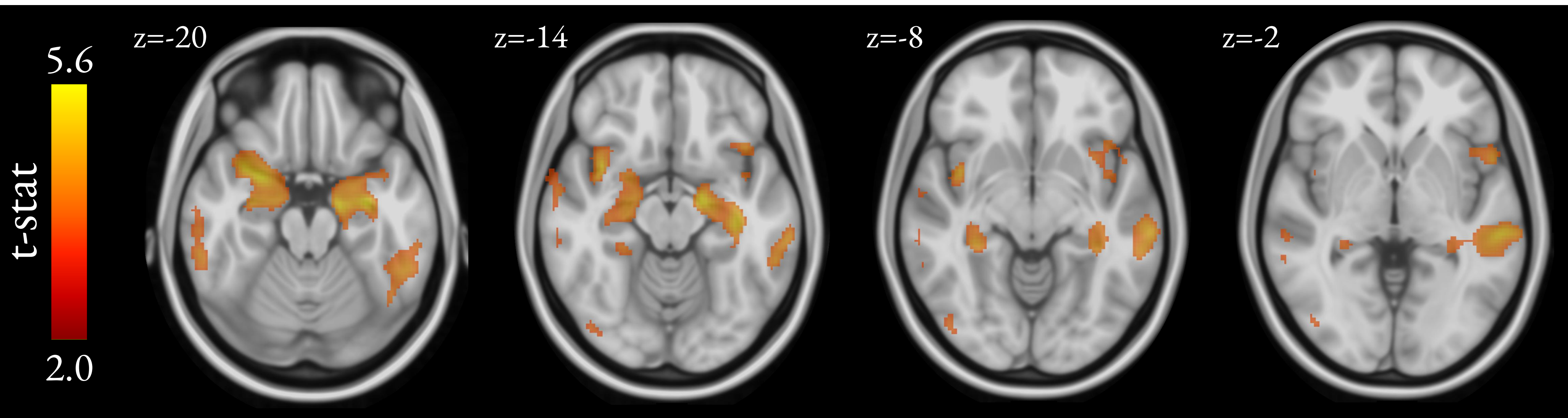


Fig 3. Gray matter reduction in Alzheimer's disease (AD) via voxel-based morphometry (VBM). Data from the OASIS-1 dataset (CDR: Clinical Dementia Rating). Controls (n=36, CDR=0) vs. AD (n=25, CDR≥0.5); contrast: Controls > AD. Axial slices show bilateral hippocampus and temporal cortex atrophy at 1,731 voxels after permutation testing (randomise, 5,000 iterations, TFCE, family-wise-error corrected $p < 0.05$; age and sex as covariates).

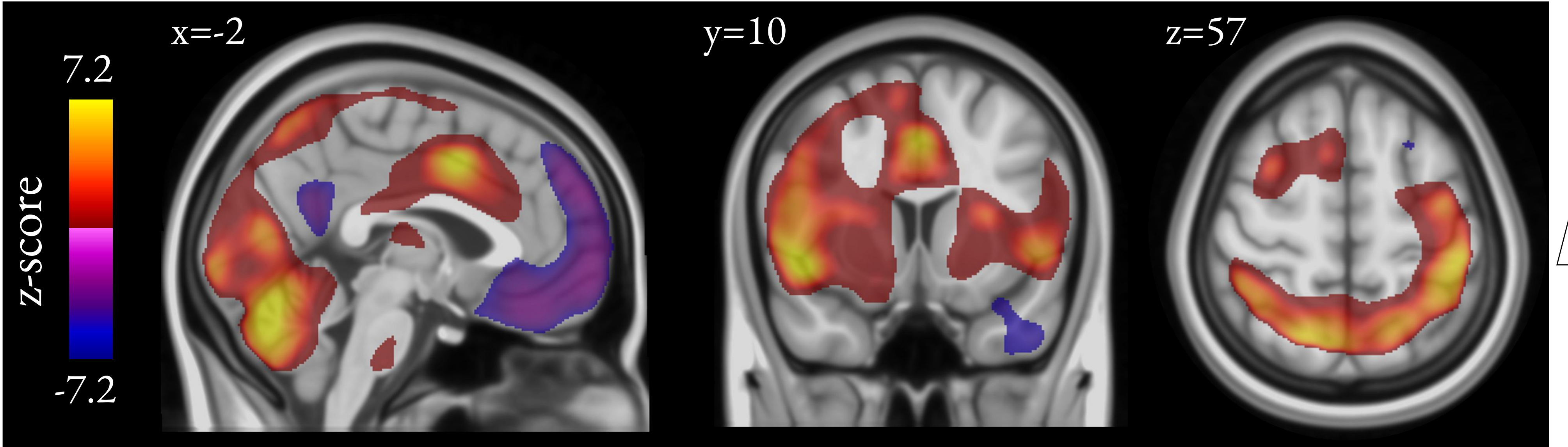
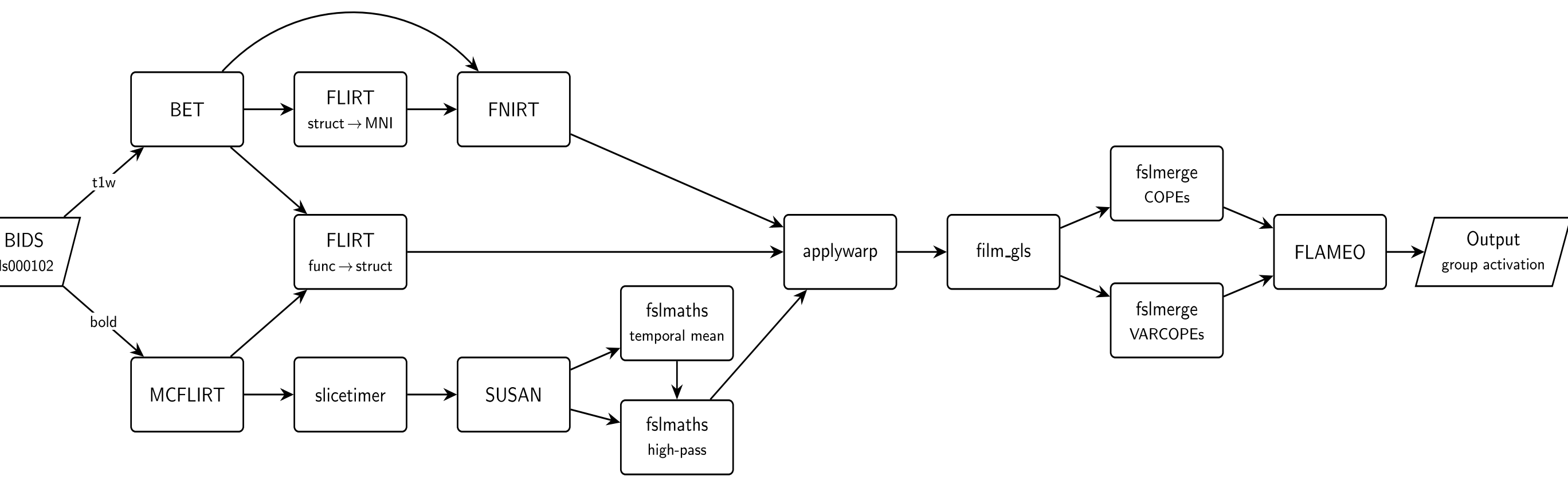
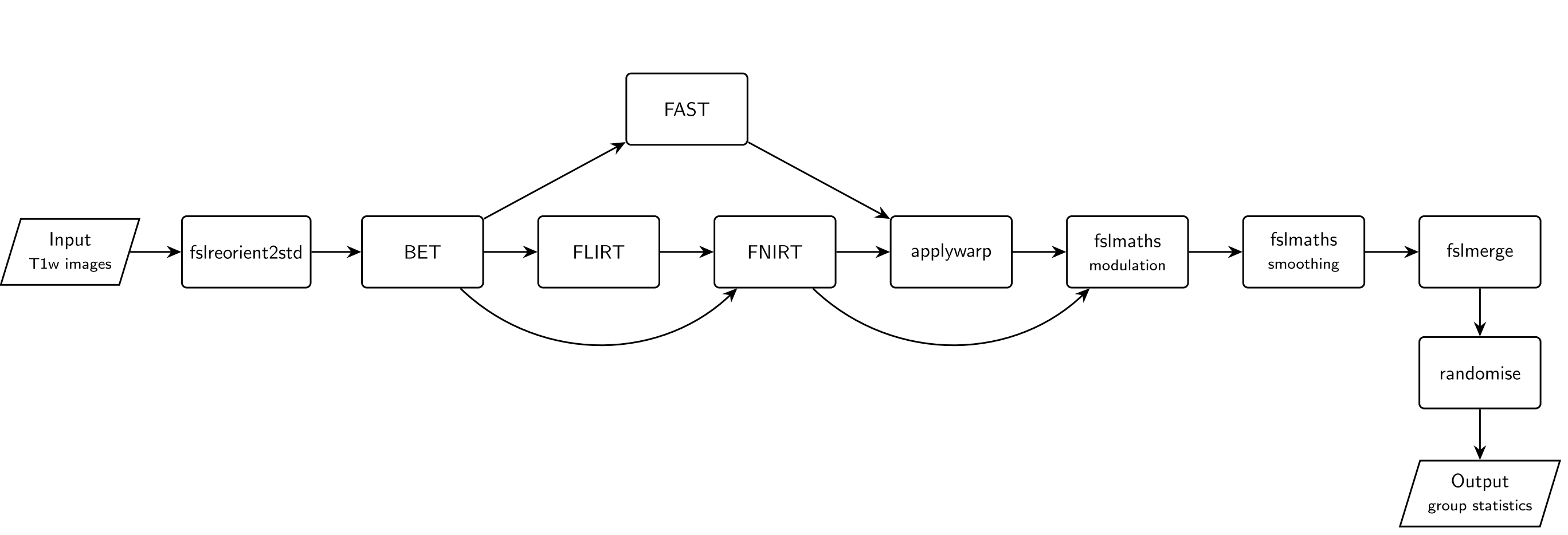


Fig 4. Cortical activation during Eriksen Flanker task. Data from OpenNeuro (ds000102, n=26). Group-level activation map for (incongruent + congruent) vs. baseline shows activation of anterior cingulate and occipital cortices with task-related deactivation in prefrontal regions, following group-level mixed effects analysis (FLAMEO; cluster-forming $z > 3.1$).



Full workflow and instructions for reproducing these analyses are available in the documentation.